Variable Acceleration 1

Mechanics Year 1 · Chapter 11

Variable Acceleration

Functions of time

Differentiation and turning points

Integration and distance travelled

1

Functions of Time 2

1. A particle has velocity $v = \frac{1}{2}t^3$ m/s. Find its velocity after 4 s, and the time when $v = 108$ m/s.
2. Another particle has $v = 2t^2 - 16t + 24$, for $t \geq 0$. Find:
   a. initial velocity;
   b. times at rest;
   c. time when $v = 64$;
   d. greatest speed on $0 \leq t \leq 5$.

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6. A particle starts at the origin with velocity
$v = 8t - 2t^2$ m/s, $t \geq 0$.
   a. When is it next instantaneously at rest?
   b. Find its greatest speed on $0 \leq t \leq 4$.

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7. A particle leaves O and later returns to O. Its distance from O is modelled by
$s = 3t^2 - t^3, \quad 0 \leq t \leq T$.
The model is valid while $s \geq 0$. Find T and explain.

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8. A particle has velocity

$$v = \tfrac{1}{5}(3t^2 - 10t + 3), \quad t \geq 0.$$

a. Find the times when it is at rest.

b. Find its greatest speed on $0 \leq t \leq 3$.

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Using Differentiation 6

Using differentiation

$$v = \frac{dx}{dt}, \qquad a = \frac{dv}{dt} = \frac{d^2x}{dt^2}.$$

A particle has $x = t^4 - 32t + 14$ m. Find:

a. velocity at $t = 3$;

b. time at rest;

c. acceleration at $t = 1.5$.

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Test Your Understanding 7

A cat's displacement from a house is

$s = t^3 - \frac{3}{2}t^2 - 36t$ m, $\quad t \geq 0$.

a. Find its velocity at $t = 2$.

b. When is it at rest?

c. Find its acceleration at $t = 5$.

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5. A particle has velocity $v = kt - 3t^2$ m/s. Its initial acceleration is 4 m/s^2.

   a. Find k.

   b. Find its acceleration when it is next instantaneously at rest.

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6. A printer head has displacement

$$s = \tfrac{1}{4}(4t^3 - 15t^2 + 12t + 30) \text{ cm},$$
$$0 \le t \le 3.$$

Find the distance between the points at which it is instantaneously at rest, to 1 d.p.

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Maxima and Minima Problems 10

A yo-yo's distance below the hand is

$s = 0.6t + 0.4t^2 - 0.2t^3, \quad 0 \le t \le 3.$

a. Justify the time restriction.

b. Find the maximum distance from the hand to 3 s.f.

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Test Your Understanding 11

1. A dolphin's velocity is $v = t^3 - 16t^2 + 64t$ m/s until the first maximum, after which it maintains that velocity. Find the time and maximum velocity.
2. A particle has $v = 2t^2 - 14t + 20$, $t \geq 0$. Find its times at rest and greatest speed on $0 \leq t \leq 4$.

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Exercise 11C – Mech Y1 Page 187-188 12

5. A particle starts at O; its distance from O is $s = \frac{9}{2}t^2 - t^3, \quad 0 \leq t \leq 4.5$.
   a. Sketch its displacement–time graph.
   b. Justify the restriction.
   c. Find its greatest distance from O.
   d. Find the acceleration magnitude there.

Sketch

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6. A train moves along a straight 4 km test track. Its displacement from the start is modelled by

$s = 3.6t + 1.76t^2 - 0.02t^3.$

$0 \le t \le 90.$

Show that it never reaches the far end of the track.

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Using Integration 14

Using integration

$$v = \int a\,dt, \qquad x = \int v\,dt.$$

Use initial conditions to find each constant.

A particle has $v = 6t - t^2$, $t \ge 0$, and $x(0) = 5$.

a. Find $x(t)$.

b. Find its distance from its starting position at $t = 6$.

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Further Example 15

A particle has velocity $v = 5 - 3t^2$ m/s, $t \geq 0$. Find the distance travelled in the third second of its motion.

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Further Example 16

A particle has $v = 2t^2 - 9t + 4$ m/s and $x(0) = 15$ m.

a. Find the times at rest.

b. Find acceleration at $t = 5$.

c. Find total distance travelled on $0 \leq t \leq 5$.

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8. A particle starts at O with velocity

$$v = 6t^2 - 26t + 15 \text{ m/s}.$$

Find the two non-zero times when it passes through O.

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9. A particle starts at O with velocity

$$v = 3t^2 - 12t + 5 \text{ m/s}.$$

a. When does it return to O?

b. Find the distance travelled on $2 \leq t \leq 3$.

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10. A particle has acceleration $a = 4t - 3$ m/s^2 and initial velocity 4 m/s. At $t = T > 0$, its velocity is again 4 m/s. Find T.

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Constant acceleration formulae 20

Constant acceleration formulae

A particle has constant acceleration a, initial velocity u, and initial displacement zero. Use integration to prove:

a. $v = u + at$;

b. $s = ut + \frac{1}{2}at^2$.

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3. Prove that $s = ut + \frac{1}{2}at^2$, with u, a constant, describes constant acceleration a.
4. Which displacement functions have constant acceleration?

A. $s = 2t^2 - t^3$

B. $s = 4t + 7$

C. $s = t^2/4$

D. $s = 3t - 2/t^2$

E. $s = 6$

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6. A train passes A at $t = 0$ and B at $t = 40$. Its distance from A is

   $s = 25t - 0.2t^2, \quad 0 \le t \le 40.$

   a. Find AB.

   b. Show acceleration is constant.

   c. At $t = 0$, a bird passes B towards A at 7 m/s, with constant acceleration 0.6 m/s² towards A. How far from A is it when directly above the train?

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